Sujet Phodapol

ROBOTICS MASTER STUDENT

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Education

KTH Royal Institute of Technology *Stockholm, Sweden*

MSc Systems, Control and Robotics *Aug 2021 - Present*

- KTH One-Year Scholarship 2022, awarded to master's students who have achieved outstanding results in their first year of studies.
- **Thesis:** *"Distributed Predictive Controllers for Load Transportation in Microgravity Environments"*
- **Courses:** Control Theory and Practice, Advanced Course, Artificial Intelligence and Multi Agent Systems, Hybrid and Embedded Control Systems, Applied Estimation, Image Analysis and Computer Vision, Introduction to Robotics, Modelling of Dynamical Systems

ETH Zürich *Zürich, Switzerland*

MSc Robotics, Systems and Control (Exchange) *Sep 2022 - Feb 2023*

- **Semester Projects:**
  - *"Data-driven adaptive control: a geometric approach"*, Automatic Control Laboratory
  - *"Design Optimization for Serial Elastic Actuators on Quadrupedal Robots"*, Robotic Systems Lab
- **Courses:** Advanced Model predictive control

Chulalongkorn University *Bangkok, Thailand*

B.Eng. in Mechanical Engineering, GPA 3.96/4.00 (First-Class Honors, Gold Medal) *Aug 2016 - May 2020*

- **Thesis:** *"An Agile Quadruped Robot"*, Designed and prototyped a quadruped robot with 8 degrees of freedom. Series elastic actuators (SEA) are used for compliance behaviour. A robot used feedback control to perform locomotion in two different gaits: walk and trot gait.

Work Experience

KTH Royal Institute of Technology *Stockholm, Sweden*

Research Engineer at DISCOWER Space Robotics Laboratory *Jun 2022 - Aug 2022*

- **Air-Carriages for Microgravity Testbeds**: Design and develop a testing platform for autonomous systems that operate in microgravity or weightless environments. These air carriages will be used to test multiple systems (holonomic and non-holonomic) in the Space Robotics laboratory. The platforms can create an air cushion between the air bearing and a flat surface using a pneumatic system. Thus, the robot can operate with virtually no friction with the operating surface.
  project website: `https://wasp-sweden.org/nest-project-discower`
- Awarded the best presentation in the final presentations of the Digital Futures Summer Internship Programme 2022.

Vidyasirimedhi Institute of Science and Technology (VISTEC) *Bangkok, Thailand*

Robotics Engineer *Nov 2020 - July 2021*

- Formulated a numerical model for physical simulation in MATLAB and fabricated a prototype for a Morphological Adaptation for Speed Control of Pipeline Inspection Gauges (MC-PIG).
- Developed the state estimation using an extended Kalman filter to predict the position and velocity of the inspection robot in C++ using ROS and CoppeliaSim.

Publications

2023 **S. Phodapol**, A. Harnkhamen, N. Asawalertsak, S. N. Gorb and P. Manoonpong, "Insect Tarsus-Inspired Compliant Robotic Gripper With Soft Adhesive Pads for Versatile and Stable Object Grasping", (RA-L 2023)

2022 **S. Phodapol**, T. Chuthong, B. Leung, A. Srisuchinnawong, P. Manoonpong, N. Dilokthanakul, "GRAB: GRAdient-Based Shape-Adaptive Compliant Locomotion Control", (RA-L and ICRA 2022)

2021 **S. Phodapol**, T. Suthisomboon, P. Kosanunt, R. Vongasemjit, P. Janbanjong, P. Manoonpong (2021), "Morphological Adaptation for Speed Control of Pipeline Inspection Gauges (MC-PIG)", (ADIPEC 2021)

2021 Manoonpong, P., **Phodapol, S.**, Suthisomboon, T., Kosanunt, P., Kasemwarapach, T., Janbanjong, P., "A Device for Moving in a Pipeline". Petty Patent Filing No: 2103002124, (Thai Patent Office)

University Projects

Multi-agent system for soccer game *Stockholm, Sweden*

KTH Royal Institute of Technology *Apr 2022*

- Developed and implement an algorithm in C for 3 vs 3 soccer games (holonomic and nonholonomic system) in Unity based on behaviour tree.

Behavior tree for service robot (TIAGo robot) *Stockholm, Sweden*

KTH Royal Institute of Technology *Oct 2021*

- Developed a behaviour tree to automate the transporting task for a service robot, including localization and navigation in ROS.

Automated storage and retrieval system *Bangkok, Thailand*

Chulalongkorn University *Nov - Dec 2019*

- Developed the prototype of industrial transportation system using Raspberry Pi.
- Created an IoT cloud platform to communicate between a robot and control room and built localization system from bird's eye view camera to control the robot's position

Smart Camera *Bangkok, Thailand*

Chulalongkorn University *Nov - Dec 2019*

- Implemented face recognition using deep learning library and SVM model in python.
- Built a 2-DOFs security camera to track anonymous people and send data to cloud using Raspberry Pi and NETPIE (IoT cloud platform)

3D-printed Quadruped robot *Bangkok, Thailand*

Chulalongkorn University *Aug - Sep 2019*

- Developed a motion sensor controller to control a multi-legged robot using STM32.
- Created quadrupedal robot using 3D printing and remote controlled via Bluetooth

Skills

Programming	Python, MATLAB, ROS, C/C++
CAD/CAM	Fusion360, CATIA V5R21, Solidworks, AutoCAD
Simulation	ANSYS, Simulink (MATLAB), Gazebo, Coppeliasim
Embedded system	Arduino, STM32, Raspberry Pi

Languages

English	Professional proficiency
Thai	Native proficiency